Koblitz N83 baseline in the requested subgroup range

Source reconciliation · 26 October 2026 · no new trial or performance claim

The 16 October ICPERF assessment identified its local GF(2^53) comparator as a 45-bit subgroup. The current cryptanalysis baseline ledger also has Koblitz N83 on curve `EC1N83Ce1h35c7ced6b497`, with subgroup order `8,569,786,107,849,059` (52.93 bits). This is an in-range reference, but the measured cold cost excludes factor-base construction. The full three-family, arity-2-through-8 comparison remains open.

Current source row

The ledger at cryptanalysis main commit `e7cf096b4e2f9020e1326e4ed0b5ac0e96cc99f9` names rank-finder run `39527997793` as the N83 baseline. It reports 12 new public points, 36 alternating predecessor/current pairs, same-point tuned rho, and 282 valid rows. The source's paired cold ratio is 1.0196 with a 95% interval of [1.0147, 1.0255]. Its A/A cold control is 0.9982 [0.9928, 1.0036]. This PDF has not independently replayed those receipts.

The same run reports normalized IC S=0.4466 and rho S=0.1138, a point IC/rho ratio of 3.9242. No interval was computed for that ratio in the source run. The fixed recipe uses a K600 orbit base with a PDP4/S3-root-index path; it does not measure every requested arity.

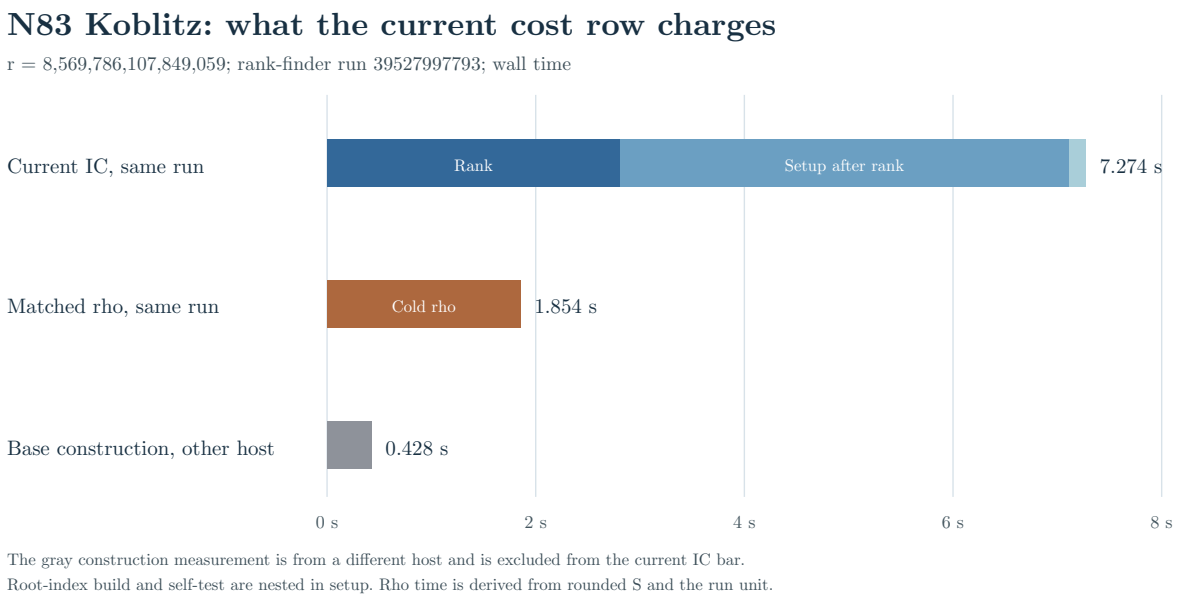

Figure 1: Source-ledger N83 phase map. Rank, setup and online are from run 39527997793; factor-base construction is from separate replay 39007026856 on another host class. The separate bar is excluded from the current cold IC total.

Charging boundary and next measurement

The current run timed 2,806.5 ms of rank work, 4,308.5 ms of subsequent setup, and 158.6 ms online. Root-index construction (about 3.25 s) and self-test (about 0.84 s) are already inside setup. The K600 factor base was loaded from a file. A different host class measured 427.524 ms of base construction, which cannot be inserted into the current run to form an accepted same-host total. The reported IC S is a lower bound on a cold total that charges nonnegative construction time.

The discriminating comparison freezes the current N83 producer and same unseen points, charges target-blind base construction through certificates and verification, and runs matched rho in the same environment. An elite orbit-base arm needs its own construction cost and a direct pairing against this producer; an earlier K899 result used a different predecessor. Prime and ordinary-binary complete rows and m=2..8 remain separate obligations.

Source

Cryptanalysis `docs/ic/BASELINES.md` (Git blob `4a444fac7f5c61992559a57601ee03c48cddab55`) and `baselines.json` (Git blob `8cfddf76272ab0ffb3f5d3c71bdad1e0cba4928e`) at the pinned commit above. Full links, distinctions and limitations are in `README.md` beside this PDF. No scientific status or claim record changes here.